THETA GATE

An autonomous options agent that cannot place a trade it should not.

The model proposes a direction. Deterministic Python does everything else. A pure-function risk guard has the last word, and cannot be argued with.

ALPACA AI TRADING AGENTS HACKATHON • 28 AUG -- 4 SEP 2026 • PAPER TRADING

The tension nobody else had to resolve

THE PROBLEM

Alpaca shipped their paper-trading skills on **25 August** — the day before this build started. Their own guidance requires a **human confirmation before every order**, and five operations demand it *regardless* of the unattended-mode setting:

`cancel_all_orders` `close_all_positions` `close_position`
`exercise_options_position` `do_not_exercise_options_position`

That is fundamentally incompatible with the autonomous agent this hackathon asks for. A cron at 10:30 on a Tuesday has nobody to ask.

“Every deployed path asserts paper at startup… There is no operator watching to catch a wrong endpoint, and a live account returns the same response shape as a paper one.” — Alpaca’s own skill

Confirmation is two things wearing one name

THE IDEA

A human confirmation does **legibility** (showing what is about to happen) and **authority** (deciding whether it may).

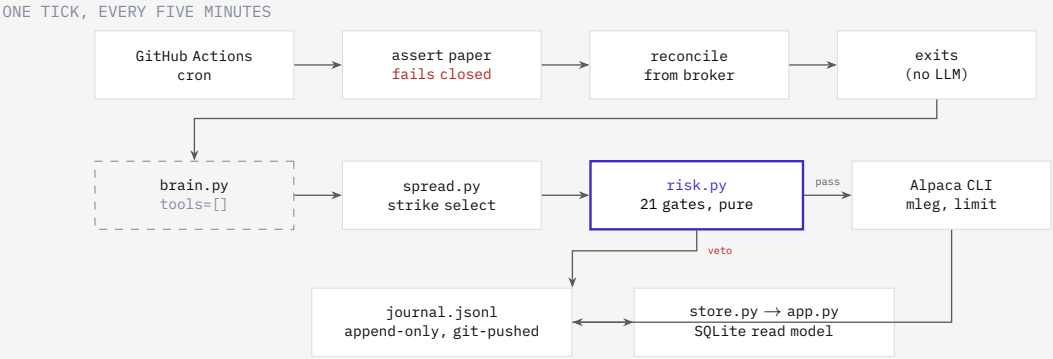
Alpaca automates away the second whenever a human is absent, and replaces it with an assertion that fails closed. **Theta Gate does the same thing, with 21 assertions instead of one.**

The one idea: the LLM has no tools at all. Every write goes through deterministic Python. The proposer suggests a direction; it never picks a strike, sizes a position, or holds an order credential.

The breach is *not reachable* — not merely forbidden by a prompt.

This is why the guard is pure functions with no I/O: it is the only component in the system holding a real decision.

Architecture



The dashed box is the only non-deterministic component in the system, and it is the one with no network access, no credential, and no ability to write anything.

The LLM boundary, in full

BRAIN.PY

```
options = ClaudeAgentOptions(
    system_prompt=SYSTEM_PROMPT, model=MODEL_ID, max_turns=1,
    tools=[], allowed_tools=[], mcp_servers={},
    strict_mcp_config=True, setting_sources=[],
)
```

Zero tools. Zero MCP servers. One turn. No filesystem, user, or project settings. It sees only the same scalar market numbers already computed for the gates — no chain, no news, no broker credential.

It may return exactly five fields: `underlying`, `direction`, `confidence`, `thesis`, `invalidation`.

Why not a read-only allowlist? An allowlist still holds a network client and still depends on being maintained correctly forever. `tools=[]` cannot reach anything, and `strict_mcp_config=True` means a stray `.mcp.json` cannot quietly re-arm it.

Any failure — malformed JSON, schema violation, timeout, or a thesis that reads back an injected instruction — produces no proposal, never a crash.

The guard

RISK.PY --- PURE FUNCTIONS, NO I/O, TIME INJECTED

ENVIRONMENT

- paper asserted before *every* order
- kill-switch file; account status; options level ≥ 3

REGIME (entry-only, never blocks an exit)

- VIX < 30 and VIX9D $<$ VIX3M
- |intraday move| $< 2\%$
- FOMC / CPI / PCE / NFP blackout

CONTRACT

- both legs need delta *and* IV — the 0DTE guard
- $6 \leq$ DTE ≤ 9; short leg $0.16 \leq |\delta| \leq 0.25$
- quote age $\leq$ 60s; spread $\leq 15\%$ of mid

PRICE

- credit within $\pm 40\%$ of $0.8\times$ short delta
- absolute floor: credit $\geq 10\%$ of width
- ATM IV $-$ 20d realised vol $\geq$ 2.0 points

SIZE AND EXPOSURE

- max loss $\leq$ \$1,000; open risk $\leq$ \$3,000
- ≤ 2 open, ≤ 1 per underlying
- buying power $\geq$ \$25,000 *and* $\geq 5\times$ max loss

DRAWDOWN

- -1% on the day $\rightarrow$ no new entries
- equity $\leq$ \$98,000 $\rightarrow$ HALT and close out

First rejection wins and is final. No gate is re-evaluated after a veto, and no LLM sits downstream of one. Every number lives in `governance.json`, rendered verbatim on the dashboard beside the line *"no LLM can write to this file"*.

The strategy, and its honest edge

WHAT IT ACTUALLY TRADES

Short credit vertical on SPY or QQQ. $5 wide. 6–9 DTE. Short leg at 0.16–0.25 delta. Exactly one contract, all week.

WHY TWO LEGS, NEVER A CONDOR

Half the legs, half the fills to win. Roughly 10% of eligible paper fills come back partial, and a 4-leg order that half-fills *manufactures a naked short*. Non-negotiable.

WHY NOT 0DTE

0DTE has no greeks and no IV, so a delta-targeting rule has no input at all. This is structural, not a preference.

There is no arithmetic edge, and we say so.

Sweeping every strike from 0.15 to 0.45 delta across five widths, expected value is **negative in every case** — between $-\$4$ and $-\$8$ per contract, precisely the bid-ask cost. Delta *is* the risk-neutral probability, so a fairly priced chain cannot yield edge by arithmetic.

The only defensible claim is the **variance risk premium**: implied vol has historically exceeded realised. Real, thin, and **six sessions cannot measure it**.

The LLM is a direction filter and a veto. It is not the edge.

Three things live testing corrected

ONE REAL SPREAD, OPENED AND CLOSED, 26 AUGUST

1. The original credit gate would have vetoed every trade. We specified credit/width of 0.20–0.45. Observed on SPY at 765.55, 7 DTE, short leg at -0.197 delta: net credit **0.60 — a ratio of 0.120**. Widening makes it *worse*, monotonically. Across the chain the real relationship is **credit/width** $\approx 0.8\times$ **short delta**. Recalibrated to a relative test.

2. Margin held is the full width, not the max loss. The fill consumed exactly **$500** of options buying power on one $5-wide contract — width $\times$ 100 $\times$ qty — while max loss was $443. Two different numbers; the buying-power gate used the wrong one.

3. A duplicate `client_order_id` is rejected, never duplicated. Alpaca returns HTTP 422. That single verified fact is the entire idempotency mechanism — it is what lets a crashed tick re-run safely with *no database at all*.

Every one of these was a plan that read fine and was wrong.

Durability without a database

WHAT WE DELIBERATELY DID NOT BUILD

The canonical design called for Postgres with row-level security, six credentialed roles, OIDC/Sigstore attestation, fenced distributed leases and hash-chained event sourcing. That is weeks of distributed-systems work. **We cut it, on the record, and said why.**

WHAT REPLACED IT

- **Idempotency from the broker.** A deterministic `client_order_id`, always looked up before submitting.
- **Concurrency** is one Actions `concurrency:` group — a platform feature.
- **The broker is the source of truth**, refetched every tick.
- **The journal** is append-only JSONL, written *before* the network call that might submit.

SQLite, but only downstream.

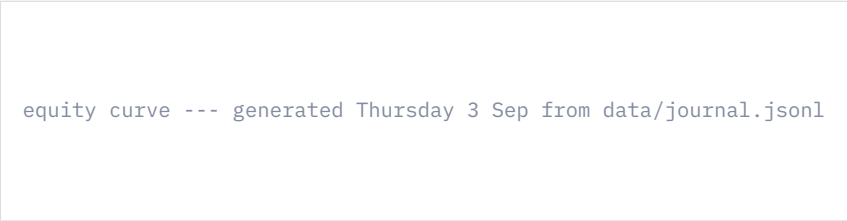
The dashboard needs aggregates, so `store.py` replays the journal into SQLite with a `sha256` hash chain, making an edited history detectable.

It is a pure *function* of the journal — rebuilt from line 1, gitignored, never authoritative. A binary `.db` in the git-publish path would collide with the one durability mechanism we actually have. JSONL merges; a database file does not.

What the agent did

RESULTS --- PAPER ACCOUNT, FRESH, $100,000

REALISED P&L	TRADES CLOSED	WIN RATE	MAX DRAWDOWN
[P&L]	[n]	[%]	[%]



Every trade above was placed by the agent. The account has never received a manual order — not one, not even a test. Its history *is* the evidence of autonomy, so every live experiment in this deck used a separate throwaway account.

What six sessions can and cannot show

THE PART MOST DECKS LEAVE OUT

The sample is **[n]** trades across six sessions. That is statistically nothing, and we are not going to claim otherwise.

WHAT THIS CANNOT DEMONSTRATE

- That the strategy has edge. At -0.197 delta the breakeven win rate is 88% against a risk-neutral 80%.
- The sign of the P&L. One 2% adverse gap inside six sessions erases the book.

WHAT IT CAN DEMONSTRATE

- That the guard held: which gates fired, how often, and that no trade was placed outside them.
- That loss is capped by construction — max loss is the width, fixed at entry. A dropped cron run cannot exceed it.
- That the agent ran unattended, reconstructable from a committed, hash-chained journal.

The number worth judging is not the P&L. It is whether the risk guard held — and that is measurable at n = **[n]**, whereas edge is not.

Built, and what it cost

ENGINEERING

THE SYSTEM

- `risk.py` — 21 gates, pure functions, no I/O, time injected
- `spread.py` — deterministic strike selection and mleg construction
- `loop.py` — one tick; recovery, reconcile, exits, entry, journal
- `brain.py` — the single bounded model call
- `store.py` / `app.py` — read model and live dashboard

THE INTEGRATION

Alpaca CLI for every broker call. No `alpaca-py`, by choice.

Nothing here is claimed untested.

Where a fact could be verified against the live API, it was — the 422 on a duplicate order id, the margin arithmetic, the credit/delta curve, the `alpaca doctor --profile` flag that silently lies.

An adversarial review pass over `loop.py` caught six real defects, two of them blockers, before any of it was trusted with an order.

The honest failure mode: roughly 4:1 against — about eight $100 wins per one $800 loss. The gates cap the left tail; they do not change the sign of a week.

THETA GATE

The risk guard is what replaces the human confirmation step.

It is the thing you must build if you want autonomy without recklessness — and it is the gap Alpaca's own tooling leaves open.

PAPER TRADING ONLY • NO MANUAL ORDERS, EVER • FULL JOURNAL COMMITTED TO GIT